

Analyses: Classification with KNN

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Description

k-nearest neighbor analyses of the feature vectors per character sequence (loaded from `~/results/features_combined.csv`). The results are stored in `~/results/classifier/`.

Load Packages

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
library(class)
library(gmodels)
library(caret)
```

Load Data

Load data table with estimated feature values per sequence.

```
# load feature value estimations
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.7295740   1.3854133 0.33333333
## 2 Aurignacien      vvvvv         5  0.0000000   0.5711903 0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047 0.22222222
## 4 Aurignacien vvvvvvvvvvvvvvv      14 0.0000000   0.6379550 0.07142857
## 5 Aurignacien      |||           3  0.0000000   0.4308271 0.33333333
## 6 Aurignacien      |||           4  0.0000000   0.5070802 0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Select subcorpora for binary classification.

```
# choose subcorpora to compare: "Aurignacien", "TeDDi", "Cuneiform (Uruk V)",
# "Cuneiform (Uruk IV)", "Cuneiform (Uruk III)".
subcorpus1 <- "Cuneiform (Uruk III)"
subcorpus2 <- "Cuneiform (Uruk IV)"
```

```
selected <- c(subcorpus1, subcorpus2)
estimations.subset <- estimations.df[estimations.df$subcorpus %in% selected, ]
```

Remove NAs (whole row)

```
estimations.subset <- na.omit(estimations.subset)
```

Center and scale the data

```
# first three columns are meta information, the others are the feature values
# be aware that this needs to be changed in case of other table format
estimations.scaled <- cbind(estimations.subset[, c("subcorpus", "sequence", "num.char")],
                           scale(estimations.subset[, c("huni.chars", "hrate.chars",
                                                         "ttr.chars", "rm.chars")]))
```

Create Training and Test Sets

```
# Generating seed
set.seed(1234)
# Randomly generating our training and test samples with a respective ratio of 2/3 and 1/3
datasample <- sample(2, nrow(estimations.scaled), replace = TRUE, prob = c(0.67, 0.33))

# Generate training set (select rows randomly by using datasample vector)
train <- estimations.scaled[datasample == 1, 1:ncol(estimations.scaled)]
# Generate test set
test <- estimations.scaled[datasample == 2, 1:ncol(estimations.scaled)]
# number of data points in test set
test.n <- nrow(test)
```

Initialize data frame

```
knn.results <- data.frame(k = numeric(0), baseline = numeric(0),
                          accuracy = numeric(0), accuracy.pvalue = numeric(0),
                          precision = numeric(0), recall = numeric(0),
                          f1 = numeric(0), test.n = numeric(0))
```

Building knn classifier

```
# choose maximum number of neighbors n
n = 10
# run a loop over different numbers of neighbors up to n
for (k in 1:n){
  # knn estimation of labels
  predictions.knn <- knn(train = train[, c("huni.chars",
                                           "hrate.chars",
```

```

        "ttr.chars",
        "rm.chars")],
    test = test[, c("huni.chars",
        "hrate.chars",
        "ttr.chars",
        "rm.chars")],
    cl = train[, "subcorpus"], k = k)

# model evaluation
# creating a dataframe from known (true) test labels
test.labels <- test[, "subcorpus"]
# combining predicted and known classes
class.comparison <- data.frame(predictions.knn, test.labels)
# giving appropriate column names
names(class.comparison) <- c("predicted", "observed")
# inspecting our results table
head(class.comparison)
# get confusion matrix
cm <- confusionMatrix(data = class.comparison$predicted,
    reference = as.factor(class.comparison$observed))
# print(cm)
# get performance metrics from the output list of confusionMatrix()
baseline <- cm$overall['AccuracyNull']
accuracy <- cm$overall['Accuracy']
accuracy.pvalue <- cm$overall['AccuracyPValue']
f1 <- cm[["byClass"]][["F1"]]
recall <- cm[["byClass"]][["Recall"]]
precision <- cm[["byClass"]][["Precision"]]

# unname and round resulting values (except for p-values)
baseline <- round(unname(baseline), 2)
accuracy <- round(unname(accuracy), 2)
f1 <- round(unname(f1), 2)
recall <- round(unname(recall), 2)
precision <- round(unname(precision), 2)

# prepare data frame with results
local.results <- data.frame(k, baseline, accuracy, accuracy.pvalue,
    precision, recall, f1, test.n,
    row.names = NULL)
# print(local.results.rounded)
knn.results <- rbind(knn.results, local.results)
# give predictions alongside the other information in data frame
knn.predictions.df <- cbind(test, predictions.knn)
}
print(knn.results)

```

```

##      k baseline accuracy accuracy.pvalue precision recall  f1 test.n
## 1    1    0.53    0.67  2.798970e-12    0.65  0.69 0.67    534
## 2    2    0.53    0.68  4.070864e-13    0.66  0.68 0.67    534
## 3    3    0.53    0.68  1.079297e-13    0.67  0.66 0.67    534
## 4    4    0.53    0.72  6.347048e-20    0.70  0.71 0.71    534
## 5    5    0.53    0.73  3.480498e-22    0.71  0.73 0.72    534
## 6    6    0.53    0.72  2.727722e-20    0.71  0.70 0.70    534

```

```
## 7 7 0.53 0.73 3.480498e-22 0.71 0.72 0.72 534
## 8 8 0.53 0.72 1.463403e-19 0.70 0.71 0.70 534
## 9 9 0.53 0.73 1.414571e-22 0.71 0.73 0.72 534
## 10 10 0.53 0.73 8.483444e-22 0.71 0.72 0.72 534
```

Write to file.

```
# write results
write.csv(knn.results, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/performance",
                                     subcorpus1, "_",
                                     subcorpus2,
                                     ".csv"
                                     ), collapse = ""),
          row.names = F)

# write df with predictions for post-hoc analyses
# note that this writes the predictions only for the last run of the loop, i.e. the highest k
write.csv(knn.predictions.df, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/predictions",
                                             subcorpus1, "_",
                                             subcorpus2,
                                             ".csv"
                                             ), collapse = ""),
          row.names = F)
```